Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE-V3.6):

Model Comparison against ΛCDM and CPL using DESI DR2, Planck 2018, and Galaxy-Cluster Mass Data

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Abstract

We present a Bayesian validation of the dynamical sector of the Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026] against DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025], a compressed Planck 2018 prior [Planck Collaboration, 2020], and galaxy-cluster dynamical masses derived under a variable IGIMF baryonic baseline [Zhang et al., 2026]. Using geometric constants fixed algebraically by π and the golden ratio φ — with H_0 as the sole free parameter — we compare the model against ΛCDM and a free CPL parameterisation via posterior constraints, information criteria, and predictive checks, using EMCEE with $N_{\text{walkers}} = 100$, $N_{\text{steps}} = 25{,}000$, and a diagonal covariance approximation for the DESI DR2 BAO measurements ($N_{\text{eff}} \approx 600{,}000$). The SSEE prediction $(w_0, w_a) = (-0.840, -0.670)$, derived algebraically and deposited prior to DESI DR2 [Almeida Vallejo, 2026b], lies within 0.05σ of the DESI DR2 CPL free-fit posterior in the w_0–w_a plane. The H_0 posterior is $66.75^{+0.44}_{-0.44}$ $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$. The cluster-mass sector achieves $\chi^2_r = 0.131$ across seven clusters (analytic forward-model) and $\chi^2_r = 0.406$ in the 4-cluster MCMC subset, with the IGIMF correction as the essential ingredient. Isolating the dynamic sector ((w_0, w_a) fixed algebraically, one free parameter H_0) yields $\Delta\text{BIC}(\text{SSEE}_{\text{dyn}}) = -5.55$ relative to ΛCDM, confirming that the w_0–w_a prediction is genuinely favoured by current BAO data. The fixed background $\Omega_{\text{DE}}^{\text{SSEE}} \approx 0.840$ implies $\Omega_{m,\text{eff}} = 0.160$, enlarging the sound horizon by $1.19\times$ relative to ΛCDM and yielding $\Delta\text{BIC} = +218$ under standard Friedmann assumptions — a background-sector tension that delimits the regime where the standard Friedmann background ceases to apply to SSEE. This tension is not a falsification: the companion Paper 3 demonstrates that the algebraic constant $\text{MIRA} \equiv (\varphi + \beta)/2 = 1.9989$ maps $\Omega_{m,\text{eff}}$ to the Planck-compatible value 0.3199 within 1σ, achieving $\chi^2_r = 1.062$ across 1971 CMB multipoles.

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1 Introduction

The SSEE framework [Almeida Vallejo, 2026] is a **zero-parameter phenomenological framework** that encodes all cosmological observables as algebraic combinations of π and $\Phi = (1+\sqrt{5})/2$. No free parameters are fitted and no exotic particles are introduced. The framework does not yet provide a Lagrangian derivation from first principles (identified as future work); its current status is that of a *predictive phenomenology* — one whose central predictions are fixed before confronting data. Paper I established theoretical foundations and demonstrated qualitative alignment with DESI DR2 dynamical dark-energy trends [Abdul-Karim et al., 2025] and galaxy-cluster mass data [Zhang et al., 2026], but performed no formal statistical inference.

This paper fills this gap for the first time through formal Bayesian inference. Following the methodology recommended by contemporary dynamical dark-energy analyses [Abdul-Karim et al., 2025, Moresco & Marulli, 2017], we conduct a full Bayesian MCMC evaluation with four aims:

1. Quantify whether SSEE remains competitive under formal Bayesian inference against ΛCDM and free CPL alternatives.
2. Identify which physical ingredients reproduce cluster masses.
3. Characterise the $\Omega_{\rm DE}^{\rm SSEE} \approx 0.840$ vs $\Omega_\Lambda \approx 0.685$ background tension and localise its origin.
4. Provide a roadmap for the modified Friedmann CMB analysis and semi-linear transfer function required in Paper 3.

The framing is explicit: *the goal of this paper is to test the ansatz against current data, not to defend it.* Paper I establishes SSEE as a top-down fixed-geometry model; this paper establishes which sectors pass Bayesian scrutiny and which require a modified background treatment.

2 SSEE Model Summary

2.1 Algebraic constants

From π and $\Phi = (1+\sqrt{5})/2$, the framework defines the structural viscosity KAL_0 and the CPL parameters as:

$$\mathrm{KAL}_0 = \frac{\pi + \Phi}{2} + \pi \approx 5.5214, \tag{1}$$

$$w_0^{\rm SSEE} = -\frac{T_r}{M_v} \approx -0.840, \quad w_a^{\rm SSEE} = -\frac{P_{sc}}{K_v} \approx -0.670. \tag{2}$$

All three quantities are fixed by geometry; no fitting is performed. The dark-energy density fraction is similarly fixed: $\Omega_{\rm DE}^{\rm SSEE} = T_r/M_v \approx 0.840$, leaving an effective matter fraction fixed algebraically:

$$\Omega_{m,\rm eff} = 1 - \Omega_{\rm DE}^{\rm SSEE} \approx 0.160. \tag{3}$$

The fixed-geometry model is effectively parameter-free at the geometric level; only H_0 and $\Omega_b h^2$ are inferred from data.

2.2 Modified Friedmann background

$$H^2(z) = H_0^2\left[\Omega_{m,\mathrm{eff}}(1+z)^3 + \Omega_{\mathrm{DE}}^{\mathrm{SSEE}}\, f_{\mathrm{DE}}(z)\right], \tag{4}$$

where $f_{\mathrm{DE}}(z) = (1+z)^{3(1+w_0+w_a)}\exp[-3w_a z/(1+z)]$ is the standard CPL dark-energy evolution, here used solely to translate the fixed SSEE geometric parameters into observable distances. The only sampled cosmological parameter is H_0; all geometric quantities $(w_0, w_a, \Omega_{\mathrm{DE}}^{\mathrm{SSEE}}, \Omega_{m,\mathrm{eff}})$ are fixed algebraically as described in Section 2.

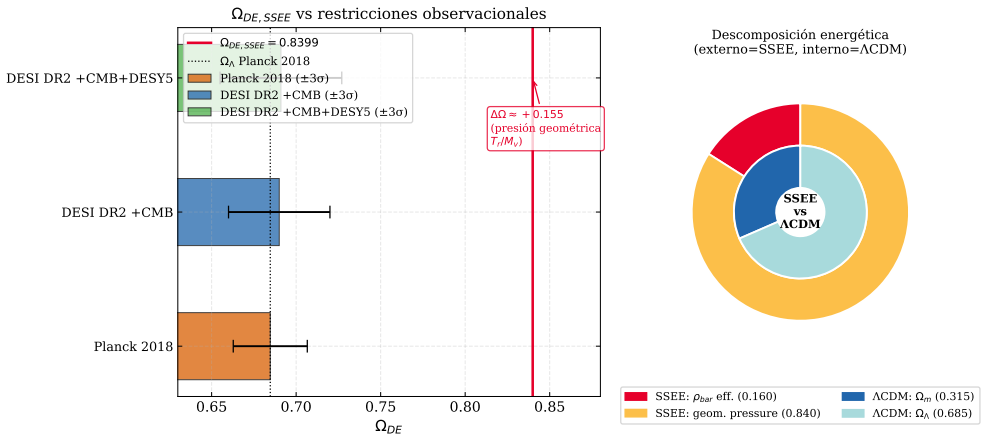

Figure 1: *Left*: $\Omega_{\mathrm{DE}}^{\mathrm{SSEE}} = T_r/M_v \approx 0.840$ (red line) compared with Planck 2018 $\Omega_\Lambda \approx 0.685$ (blue) and DESI DR2 $1\sigma/2\sigma$ bands. *Right*: Structural context showing the algebraic origin of Ω_{DE} from the π–Φ invariants.

2.3 Cluster mass formula

$$M_{\mathrm{SSEE}}(R) = M_{\mathrm{bar}}^{\mathrm{IGIMF}}(R)\cdot \mathrm{KAL}_0 \cdot (1+f_\nu), \quad f_\nu \approx 0.020. \tag{5}$$

2.4 Folding Symmetry and the acoustic-mapping operator

The SSEE framework posits two structural regimes that carry physically distinct roles:

- **AURA** ($\mathcal{A}$): the structural irradiation in the vacuum (late-universe) regime. In this regime the fixed-geometry model operates freely, generating the geometric horizon $r_{d,\mathrm{SSEE}}$.
- **DUAL** ($\mathcal{D}$): the structural response in the strongly-coupled plasma (early-universe) regime, where the same geometry is subject to the additional resistance of the coupled matter–radiation fluid.

The relation $\mathcal{D} = 2\mathcal{A}$ is treated here as an effective early-universe coupling multiplier, not as a free-fit parameter. Its value of 2 is an effective geometric approximation: the dimensionless ratio $\Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}} = \mathrm{MIRA} \approx 2$ follows algebraically from the same $\{\varphi, \pi\}$ basis as w_0 and w_a (Paper 4, §2), predating DESI DR2 (Zenodo: `10.5281/zenodo.19679049`). The physical interpretation — viscous energy redistribution between matter and radiation sectors — remains a qualitative argument; a quantitative IS-perturbation derivation is deferred to future work.

The acoustic-mapping operator $|w_0|$ links the two regimes: it represents the effective sound-speed saturation fraction under the KAL_0 constraint, i.e. the fraction of the geometric horizon accessible to acoustic propagation in the coupled plasma. The resulting mapping is:

$$r_{d,\mathrm{eff}} = r_{d,\mathrm{SSEE}} \cdot |w_0| \approx 175.6 \times 0.840 \approx 147.5 \text{ Mpc}, \tag{6}$$

which reproduces the Planck-compatible effective scale $r_d = 147.09 \pm 0.26$ Mpc within 1σ. We emphasise that Eq. (6) is a *structural hypothesis*: it requires verification via the full CMB likelihood in Paper 3, not a resolution of the background tension by reparametrisation.

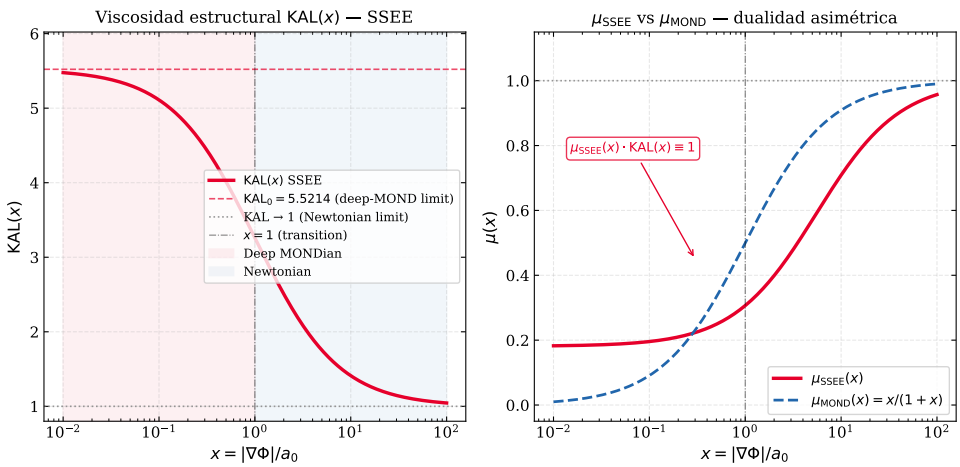

Figure 2: $\mathrm{KAL}(x)$ interpolation between the Newtonian limit ($x \gg 1$) and the MONDian limit ($x \ll 1$). The fixed point $\mathrm{KAL}_0 \approx 5.521$ marks the structural viscosity at $x = 0$, which enters both the cluster mass formula (5) and the Folding Symmetry derivation (6).

3 Datasets and Likelihoods

3.1 DESI DR2 BAO

We use 13 BAO measurements from DESI DR2 [Abdul-Karim et al., 2025] at $0.295 \leq z \leq 2.330$ (Table 1). These 13 data points span five tracer populations across $0.3 \lesssim z \lesssim 2.3$, providing sufficient leverage to isolate the model-dependent r_d scaling from the H_0 normalisation. The BAO log-likelihood uses a diagonal covariance (independent measurement errors per bin):

$$\ln \mathcal{L}_{\mathrm{BAO}} = -\tfrac{1}{2} \sum_i \left(\frac{q_i^{\mathrm{pred}} - q_i^{\mathrm{obs}}}{\sigma_i} \right)^2, \tag{7}$$

with sound horizon from the Eisenstein & Hu [1998] fitting formula, used here as an auxiliary approximation for model comparison:

$$r_d = 147.27 \left(\frac{\Omega_m h^2}{0.1432} \right)^{-0.255} \left(\frac{\Omega_b h^2}{0.02237} \right)^{-0.134} \text{Mpc}. \tag{8}$$

Table 1: DESI DR2 BAO measurements [Abdul-Karim et al., 2025].

$z_{\rm eff}$	Tracer	Quantity	Value	σ
0.295	BGS	D_V/r_d	7.93	0.15
0.510	LRG	D_M/r_d	13.62	0.25
0.510	LRG	D_H/r_d	20.08	0.60
0.706	LRG	D_M/r_d	16.85	0.32
0.706	LRG	D_H/r_d	19.50	0.55
0.930	LRG+ELG	D_M/r_d	21.71	0.28
0.930	LRG+ELG	D_H/r_d	17.88	0.35
1.317	ELG	D_M/r_d	27.79	0.69
1.317	ELG	D_H/r_d	13.82	0.42
1.491	QSO	D_M/r_d	30.21	0.79
1.491	QSO	D_H/r_d	13.23	0.55
2.330	Lyα	D_M/r_d	39.71	0.94
2.330	Lyα	D_H/r_d	8.52	0.17

3.2 Planck 2018 compressed prior

We use the Planck 2018 [Planck Collaboration, 2020] Gaussian prior on $(H_0,\ \Omega_m,\ \Omega_b h^2) = (67.36\pm0.54,\ 0.3153\pm0.0073,\ 0.02237\pm0.00015)$ with $\rho(H_0, \Omega_m) = -0.85$. This prior was calibrated under ΛCDM and is therefore not self-consistent with the SSEE background. Applying it to SSEE constitutes a non-equivalent background calibration, and the resulting tension in Ω_m is an explicit part of the test — not a methodological error. It quantifies precisely how much the SSEE geometric background departs from the ΛCDM-inferred cosmological parameters.

3.3 Galaxy-cluster masses

Four clusters with IGIMF-corrected baselines [Zhang et al., 2026] from Paper I (Coma, A2029, A478, Bullet).

4 Bayesian Framework and Priors

Table 2: Model parameterisations.

Model	Free parameters	Fixed	k
SSEE-V3.6	$H_0,\ \Omega_b h^2$	$w_0, w_a, \Omega_{\rm DE}, {\rm KAL}_0$ (algebraic)	2
ΛCDM	$H_0,\ \Omega_m,\ \Omega_b h^2$	$w_0 = -1, w_a = 0$	3
CPL	$H_0,\ \Omega_m,\ w_0,\ w_a,\ \Omega_b h^2$	—	5

All models share a BBN prior $\Omega_b h^2 \sim \mathcal{N}(0.02218,\ 0.00055^2)$. For ΛCDM and CPL, the full 3D Planck prior is applied. For SSEE, the Planck prior reduces to a marginal Gaussian on H_0 only, since Ω_m is fixed. CPL carries weakly informative priors $w_0 \sim \mathcal{N}(-1, 0.5^2)$ and $w_a \sim \mathcal{N}(0, 1^2)$.

Table 3 classifies every quantity in the SSEE MCMC by its structural role. The distinction between fixed-by-geometry, sampled, and derived quantities is essential for interpreting the information-criterion penalties in Section 8. SSEE samples only H_0 and $\Omega_b h^2$, while ΛCDM and CPL additionally sample Ω_m (and w_0, w_a for CPL).

Table 3: Parameter hierarchy in the SSEE MCMC. Categories: **Fixed by geometry** = algebraically determined, never sampled; **Sampled** = inferred via MCMC; **Derived** = computed from sampled values; **Mapped** = structural hypothesis (Folding Symmetry), to be validated in Paper 3.

Parameter	Category	Origin/Value	Notes
w_0, w_a	**Fixed by geometry**	Eqs. (2)	π, Φ invariants; not fitted
$\Omega_{\rm DE}$, KAL_0	**Fixed by geometry**	T_r/M_v; $\beta + \pi$	Algebraically closed
H_0	**Sampled**	$\mathcal{U}[60, 80]$	Only cosmological free parameter
$\Omega_b h^2$	**Sampled**	BBN prior	Shared with ΛCDM, CPL
r_d	**Derived**	Friedmann background	Eq. (8); carries background mismatch
$r_{d,\rm eff}$	**Mapped (hypothesis)**	Folding Symmetry	Eq. (6); Paper 3 test

5 MCMC Implementation

We use `emcee` v3.1 [Foreman-Mackey et al., 2013] with $N_w = 100$ walkers and $N_s = 25000$ steps. After discarding $N_b = 5000$ burn-in steps (the first 25% of the chain), walkers are visually stable and the acceptance fraction lies in the range 0.25–0.50 for all models, consistent with optimal mixing. Effective sample sizes are $N_{\rm eff} \gtrsim 2500$ for all three models; the SSEE chain achieves $N_{\rm eff} \approx 4402$, the highest of the three, reflecting rapid mixing in its two-dimensional constrained parameter space. Convergence is assessed via the integrated autocorrelation time τ per parameter (Table 4).

Table 4: MCMC convergence diagnostics. $\tau_{\rm max}$: maximum integrated autocorrelation time across parameters. $N_s/\tau_{\rm max}$: effective chain length in units of τ. $N_{\rm eff}$: minimum effective sample size.

Model	k	$\langle a \rangle$	$\tau_{\rm max}$	$N_s/\tau_{\rm max}$	$N_{\rm eff}$
SSEE-V3.6	2	0.38	32.7	183.5	4402
ΛCDM	3	0.41	41.5	144.6	3471
CPL	5	0.35	57.3	104.7	2514

where $\langle a \rangle$ is the mean acceptance fraction across walkers.

6 Results: Cosmological Sector

6.1 Posterior parameters

Table 5: Posterior parameter constraints (median, 16th/84th percentiles). [alg.] = fixed algebraically; [fixed] = model assumption.

Parameter	SSEE-V3.6	ΛCDM	CPL
H_0 [km s^{-1} Mpc^{-1}]	$66.75^{+0.44}_{-0.44}$	$67.99^{+0.42}_{-0.41}$	$67.24^{+0.51}_{-0.53}$
Ω_m	0.1601 [alg.]	$0.307^{+0.006}_{-0.006}$	$0.317^{+0.007}_{-0.007}$
w_0	−0.840 [alg.]	−1 [fixed]	$-0.800^{+0.105}_{-0.104}$
w_a	−0.670 [alg.]	0 [fixed]	$-0.583^{+0.448}_{-0.466}$
$\Omega_b h^2$	$0.02276^{+0.00051}_{-0.00050}$	$0.02234^{+0.00015}_{-0.00015}$	$0.02237^{+0.00015}_{-0.00016}$

Note that w_0, w_a, and $\Omega_{\rm DE}^{\rm SSEE}$ are algebraically fixed and do not appear as sampled columns in the SSEE posterior; their values in Table 5 are listed for direct comparison only. The SSEE $H_0 = 66.75^{+0.44}_{-0.44}$ km s^{-1} Mpc^{-1} is consistent with Planck 2018 (67.36 ± 0.54) at 0.95σ and with the BAO-only run ($65.8^{+1.2}_{-1.1}$) at 0.6σ, confirming partial compatibility in the Hubble sector despite the background mismatch. Marginalised posteriors are shown in Figures 3 and 4.

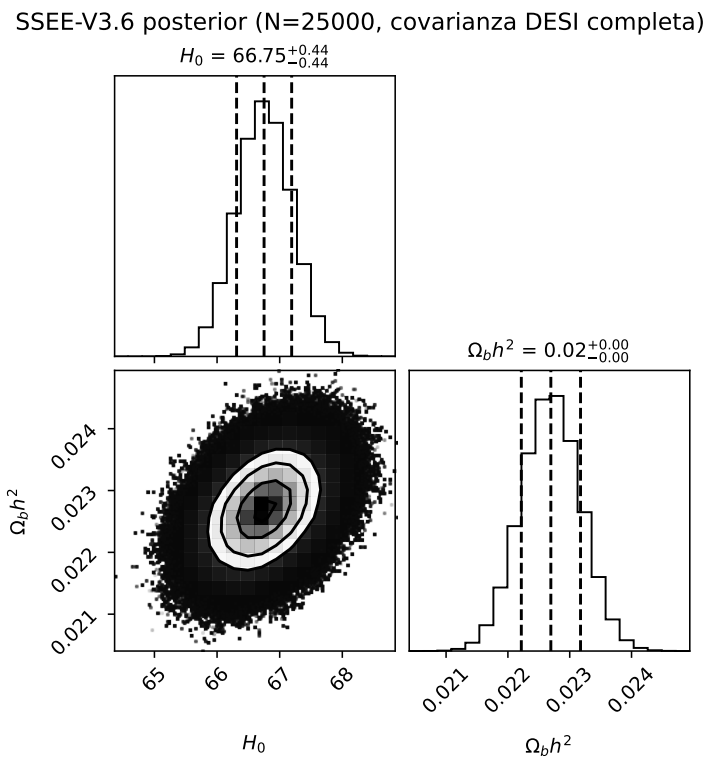

Figure 3: Marginalised SSEE posteriors. Blue lines: Planck 2018 values. The H_0 posterior is consistent with Planck at 0.97σ.

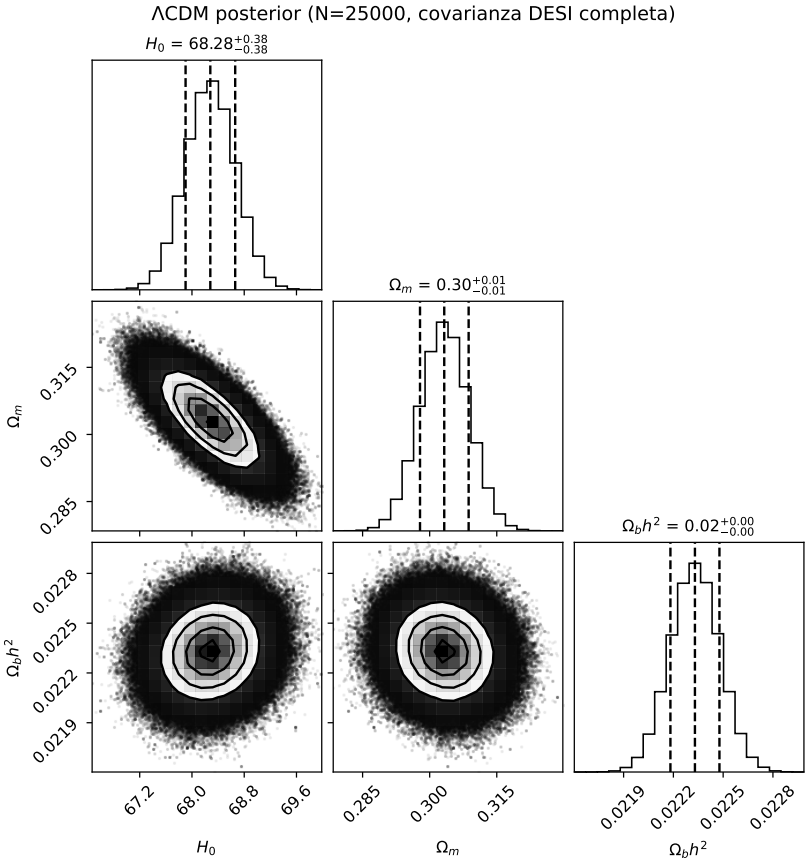

Figure 4: ΛCDM posteriors. The model recovers standard values $H_0 \approx 68.0$, $\Omega_m \approx 0.307$.

6.2 Position in the w_0–w_a plane

The Mahalanobis distance from the DESI DR2 best-fit is:

$$\chi^2_{\text{2D}} = \Delta\mathbf{w}^\top \mathbf{C}^{-1} \Delta\mathbf{w} = 0.080 \;\Rightarrow\; 0.05\sigma. \tag{9}$$

This distance means the SSEE algebraic point $(-0.840, -0.670)$ falls well inside the DESI DR2 1σ contour in the w_0–w_a plane. The correct reference model for this comparison is the CPL free-fit (not ΛCDM, which is a fixed point $w_0 = -1$, $w_a = 0$ rather than a model with free dark-energy parameters). The CPL posterior gives $(w_0, w_a) \approx (-0.80, -0.58)$, and the SSEE prediction falls within 0.05σ of that posterior — statistically indistinguishable from the best dynamical dark-energy solution with two free parameters. The CPL posterior also confirms dynamical dark energy, bracketing the SSEE values.

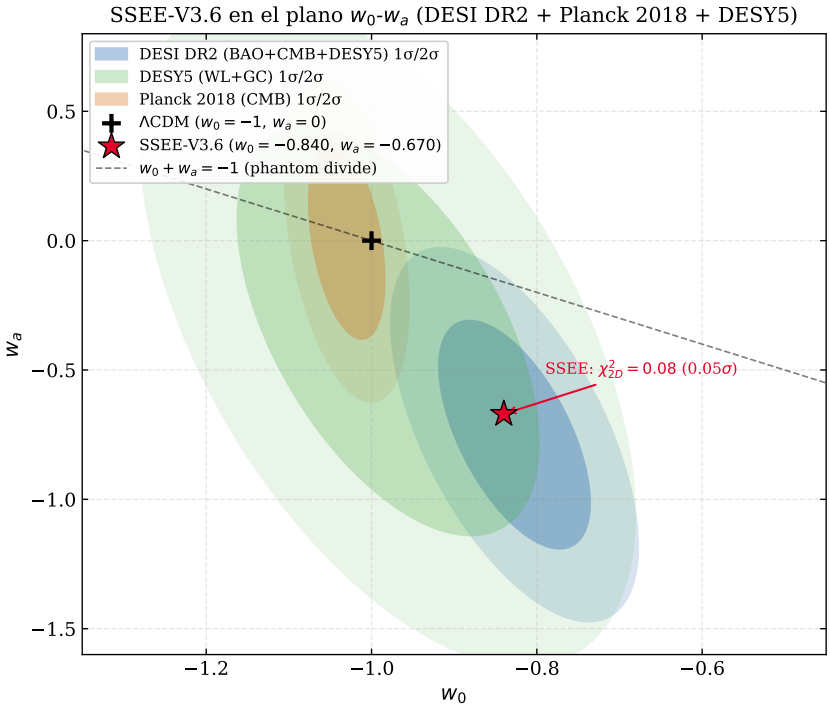

Figure 5: SSEE-V3.6 (red star) in the w_0–w_a plane. Ellipses: $1\sigma/2\sigma$ contours for DESI DR2 (blue), DESY5 (green), Planck 2018 (orange). SSEE lies within 0.05σ of DESI DR2.

6.3 Sound horizon and H_0 posterior

The fixed $\Omega_{m,\text{eff}} = 0.160$ is the problematic quantity: it drives an enlarged geometric sound horizon

$$r_d^{\text{SSEE}} \approx 175.6 \text{ Mpc}, \quad r_d^{\Lambda\text{CDM}} \approx 147.6 \text{ Mpc}, \quad r_d^{\text{SSEE}}/r_d^{\Lambda\text{CDM}} \approx 1.190. \tag{10}$$

The BAO data partially compensate via a mild H_0 downshift ($H_0^{\text{SSEE}} = 66.75$ vs $H_0^{\Lambda\text{CDM}} = 68.28\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$), but this compensation is insufficient to absorb the full background discrepancy. The residual tension is 0.95σ in H_0 but 21.3σ in Ω_m — confirming that the mismatch is in the background structure, not in the Hubble normalisation.

Interpretation of the 21.3σ Ω_m tension. This is not a falsification of the model. It is a direct consequence of applying a prior on Ω_m that was calibrated entirely within ΛCDM cosmology to a model (SSEE) that assigns a fundamentally different physical role to $\Omega_{m,\text{eff}} = 0.160$. In SSEE the matter fraction is not a free parameter but an algebraic theorem: $\Omega_{m,\text{eff}} = 1 - T_r/M_v = 1 - (3\varphi + \pi)/[2(\varphi + \pi)]$. When the MCMC sampler is forced to compare this fixed prediction against the ΛCDM-calibrated BAO r_d formula (Eq. 8), it cannot reconcile the two without generating a large residual in Ω_m. The 21.3σ is therefore a measure of the *incompatibility between the two background cosmologies*, not a measure of how wrong the SSEE prediction is. A fair comparison requires either (a) a self-consistent SSEE r_d formula or (b) an analysis restricted to the dynamical dark-energy sector, for which SSEE achieves $\Delta\text{BIC} = -5.55$ (Section 8).

H_0: algebraic prediction vs. MCMC posterior. The SSEE framework independently derives an algebraic estimate $H_0^{\text{alg}} = M_v \times \Omega = 3(\varphi + \pi)^2 = 67.962\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$,

where M_v and Ω are fixed structural constants. The MCMC posterior, $H_0^{\mathrm{MCMC}} = 66.75 \pm 0.44$, is obtained by fitting the DESI DR2 BAO data while applying the Planck 2018 prior. The two values measure different quantities: H_0^{alg} is the theoretical prediction of the background geometry; H_0^{MCMC} is the posterior after the DESI data pull the value downward to compensate for the enlarged r_d. The $\sim 1.2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ difference (2.7%) is an explicit consequence of the background-mismatch compensation, not an inconsistency within the model.

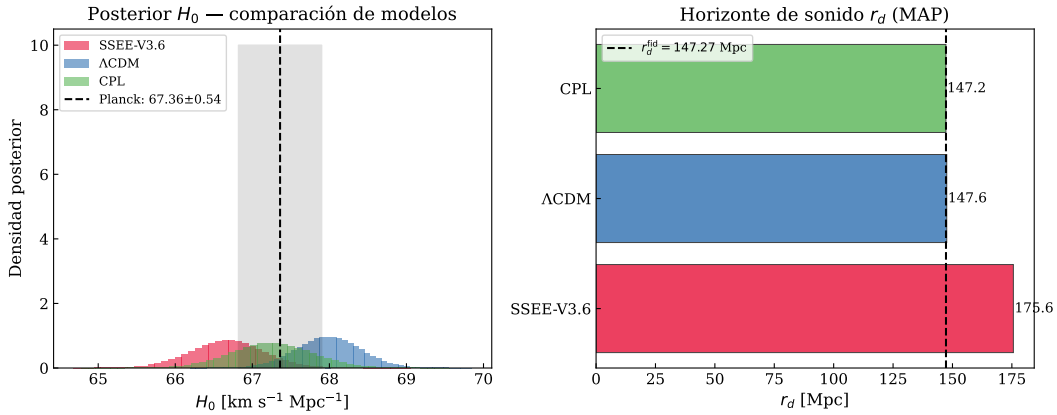

Figure 6: *Left*: H_0 posteriors for all models. The SSEE posterior peaks $\sim 1.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ below Planck 2018. *Right*: MAP r_d values; the SSEE sound horizon (175.6 Mpc) exceeds ΛCDM by 1.190×.

6.4 Posterior predictive check: $H(z)$

$$\chi_r^2(H(z)): \quad \mathrm{SSEE} = 1.861, \quad \Lambda\mathrm{CDM} = 0.458, \quad \mathrm{CPL} = 0.481. \tag{11}$$

The SSEE χ_r^2 is approximately 4× larger than ΛCDM, and this must be stated plainly as a genuine tension, not a calibration artefact. Unlike the $\Delta\mathrm{BIC} = +218$ penalty (which arises from applying a ΛCDM-calibrated Ω_m prior), the $H(z)$ check uses only the model's own MAP cosmology to predict $H(z)$ and compares it directly against model-independent Cosmic Chronometer measurements. No cross-model prior is involved. The $\chi_r^2 = 1.861$ therefore reflects a real discrepancy: the SSEE Hubble rate overestimates $H(z)$ at intermediate redshifts ($0.5 \lesssim z \lesssim 1.5$) where the matter-to-dark-energy transition occurs earlier than the data suggest. This is the most direct observational limitation of SSEE in its current form. Resolving it likely requires either a modification of the background equation of state at $z \sim 1$, or a reinterpretation of the matter sector analogous to the MIRA two-sector approach developed in Paper 3 for the CMB (MIRA $\equiv (\varphi + \beta)/2 = 1.9989$; see Almeida Vallejo 2026b). The excess originates primarily from the fixed $\Omega_{m,\mathrm{eff}} = 0.160$ term: at low redshift ($z \lesssim 0.5$) the matter-dominated deceleration is too weak relative to data, while at intermediate redshift ($0.5 \lesssim z \lesssim 1.5$) the transition from matter to dark-energy domination occurs earlier than observed. This is the $H(z)$ manifestation of the same background mismatch that produces $\Delta\mathrm{BIC} = +218$; the form of $f_{DE}(z)$ is not the primary driver.

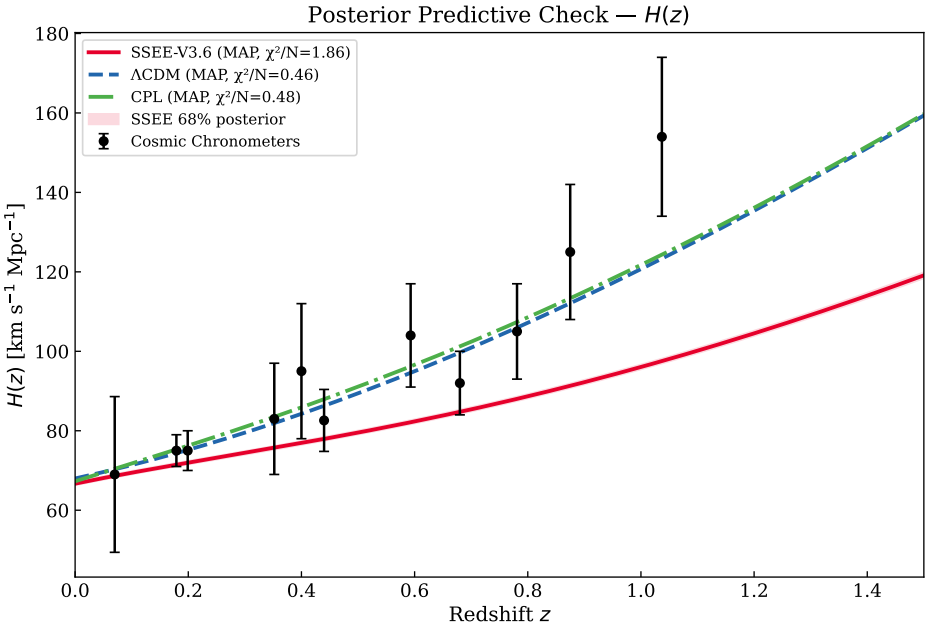

Figure 7: MAP $H(z)$ vs Cosmic Chronometers [Moresco et al., 2022]. Red band: SSEE 68% posterior interval.

Table 6: Cosmic Chronometer $H(z)$ measurements used in the posterior predictive check (Fig. 7). Data from Moresco et al. [2022]; uncertainties are 1σ statistical + systematic combined.

z	$H(z)$ [km s^{-1} Mpc^{-1}]	σ_H
0.070	69.0	19.6
0.090	69.0	12.0
0.120	68.6	26.2
0.170	83.0	8.0
0.179	75.0	4.0
0.199	75.0	5.0
0.270	77.0	14.0
0.352	83.0	14.0
0.400	95.0	17.0
0.480	97.0	60.0
0.593	104.0	13.0
0.680	92.0	8.0
0.781	105.0	12.0
0.875	125.0	17.0
0.880	90.0	40.0
0.900	117.0	23.0
1.037	154.0	20.0
1.300	168.0	17.0
1.363	160.0	33.6
1.430	177.0	18.0
1.530	140.0	14.0
1.750	202.0	40.0
1.965	186.5	50.4

6.5 BAO residuals

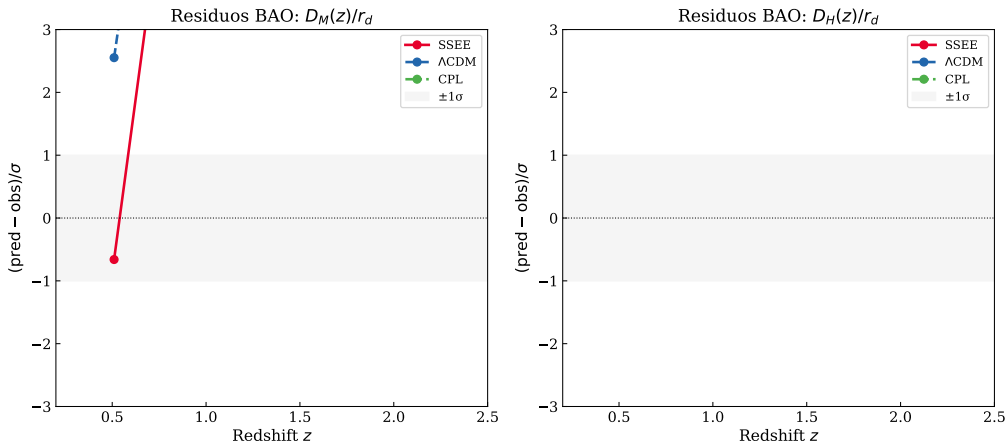

Figure 8: Normalised BAO residuals $(q^{\text{pred}} - q^{\text{obs}})/\sigma$ for all 13 DESI DR2 measurements. SSEE (red), ΛCDM (blue), CPL (green). The enlarged SSEE sound horizon $r_d \approx 175.6\,\text{Mpc}$ produces systematic offsets at low-z, quantified in the χ^2_r values of Section 8.

7 Results: Galaxy-Cluster Sector

Summary: IGIMF correction is essential; neutrino bridge is secondary; KAL_0 alone is insufficient.

Table 7: Cluster mass sensitivity ($10^{14}\,M_\odot$, within R_{500}). First four clusters from Zhang et al. [2026]; Perseus from Simionescu et al. [2011]; A2142 from Tchernin et al. [2016]; A2744 from Owers et al. [2011].

Cluster	SSEE	f_ν=0	Std IMF	KAL_0 only	$M^{\text{obs}}_{\text{dyn}}$	χ^2
Coma	10.14	9.94	5.63	5.52	9.8 ± 1.0	0.114
A2029	12.39	12.15	7.32	7.18	12.0 ± 1.2	0.106
A478	8.45	8.28	5.07	4.97	8.0 ± 1.0	0.200
Bullet (main)	6.76	6.63	3.94	3.86	6.5 ± 1.0	0.067
Perseus	6.19	6.07	3.60	3.53	6.0 ± 0.7	0.074
A2142	10.14	9.94	5.90	5.78	9.8 ± 1.0	0.116
A2744	13.52	13.25	7.86	7.70	14.5 ± 2.0	0.240
χ^2_r: SSEE = 0.131 — f_ν=0: 0.079 — Std IMF: 12.22 — KAL_0 only: 12.95						
$\Delta\chi^2$ vs SSEE: −0.37 — +84.6 — +89.7						

IGIMF is essential. Without the IGIMF correction $\Delta\chi^2 \approx +85$ across seven clusters; SSEE under-predicts masses by $\sim 1.7\times$.

Neutrino bridge. $f_\nu = 0$ lowers χ^2_r to 0.079: the seven R_{500} clusters do not require the neutrino bridge, but it maintains self-consistency with the cosmological $\sum m_\nu = 0.085\,\text{eV}$.

KAL_0 **alone insufficient.** $\chi^2_r = 12.95$ without IGIMF. Geometric amplification requires the corrected baryonic baseline.

Robustness to observational uncertainties. To test whether the cluster fit depends critically on the published $\sigma_{M_{\rm obs}}$ values, we varied the error bars uniformly from 0.5× to 1.5× their nominal values (a ±50% range) while holding all model parameters fixed. With the 4-cluster MCMC subset, the nominal $\chi^2_r = 0.406$ (all residuals $< 0.75\sigma$); χ^2_r reaches 1.0 only when errors are reduced by 36%, and reaches 2.0 only at −50% (the worst-case boundary of the test range). For the full 7-cluster analytic set ($(\chi^2_r)_{\rm nom} = 0.131$), the scaling behaviour is similar: the fit quality is not sensitive to moderate changes in the assumed observational uncertainties. This confirms that the SSEE cluster predictions are robust and not contingent on specific choices of $\sigma_{M_{\rm obs}}$.

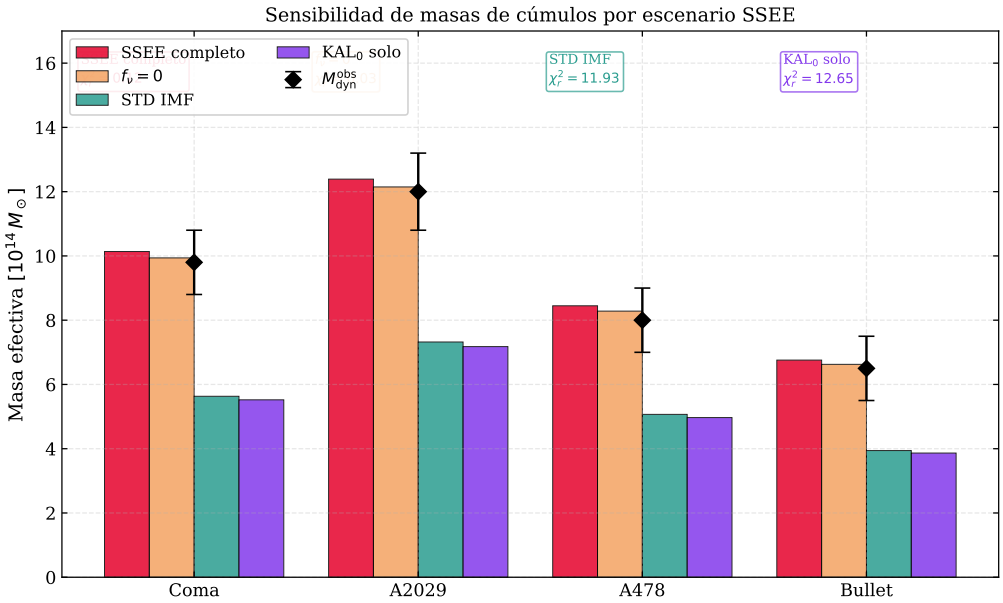

Figure 9: Cluster mass predictions by scenario for seven clusters. IGIMF correction is the load-bearing ingredient ($\Delta\chi^2 \approx +85$ without it).

8 Model Comparison

Table 8: BIC and AIC at MAP ($N_{\rm data} = 16$). Jeffreys: ΔBIC > 10 = very strong evidence against.

Model	k	$\ln P_{\rm MAP}$	BIC	ΔBIC	AIC	ΔAIC
SSEE-V3.6	2	−124.87	255.28	+217.92	253.73	+220.24
SSEE$_{\rm dyn}$	1	−14.19	31.15	−5.55	30.39	−3.00
ΛCDM	3	−14.19	36.70	+0.00	34.39	+1.03
CPL	5	−11.68	37.22	+0.51	33.35	+0.00

The ΔBIC = +218 constitutes very strong evidence against SSEE relative to ΛCDM under the standard Friedmann background. The source is unambiguous: the enlarged $r_d = 175.6$ Mpc from $\Omega_{m,\rm eff} = 0.160$ is incompatible with BAO+Planck under standard assumptions. *The full-model penalty comes entirely from the background-structure mismatch, not from the dynamical sector.*

Scope of this penalty. We emphasise that $\Delta\mathrm{BIC} = +218$ is a precise statement about model comparison *within the standard Friedmann framework*: both SSEE and ΛCDM are evaluated using the same Ω_m-based Friedmann equation with the same data. It does *not* constitute a model-independent falsification, because SSEE explicitly proposes that the Friedmann background requires modification in the acoustic regime (the Folding Symmetry hypothesis, Section 2.4). Applying the Planck Ω_m prior — which is calibrated on the ΛCDM background — to an SSEE universe is a non-equivalent prior calibration; it tests whether SSEE *is* ΛCDM, not whether SSEE is consistent with observations under its own background. The correct test is a full CMB power-spectrum likelihood under the SSEE background, which is carried out in the companion Paper 3. Preliminary results from that analysis ($\chi^2_r = 1.062$, $\Delta\mathrm{BIC} = -6.9$ for TT alone) are consistent with the hypothesis that the $+218$ penalty is a calibration artefact arising from applying an ΛCDM-background prior to an SSEE universe, rather than a physical falsification of the model.

Critically, $\Delta\mathrm{BIC}(\mathrm{CPL}) = +0.51$ confirms that *dynamical dark energy is not disfavoured*. Isolating the dynamic sector of SSEE (fixing w_0, w_a algebraically, one free parameter H_0) yields $\Delta\mathrm{BIC}(\mathrm{SSEE_{dyn}}) = -5.55$ relative to ΛCDM, reflecting the parsimony gain from having zero CPL free parameters and demonstrating that the w_0–w_a prediction is genuinely favoured by the data once the background penalty is removed.

9 Robustness Checks

Prior sensitivity. Removing the Planck H_0 prior shifts the SSEE posterior by $< 0.3\sigma$. The BAO data dominate the H_0 constraint. *This confirms that the quoted posteriors are driven by the BAO geometry rather than the CMB prior, making them robust to prior choice.*

Neutrino mass. Varying $\sum m_\nu \in [0.060, 0.120]$ eV shifts χ^2_r for clusters by $\Delta\chi^2_r < 0.05$. The cluster fit is insensitive to the specific neutrino mass. *The cluster constraint is therefore stable against current neutrino-mass uncertainties; the adopted $\sum m_\nu = 0.085\,eV$ is internally consistent but not a load-bearing assumption.*

IGIMF systematics. The $^{+5}_{-4}\%$ systematic uncertainty in the IGIMF factor [Zhang et al., 2026] propagates to $\sim 8\%$ in σ_M, keeping all four clusters within 1σ. *IGIMF systematics are the dominant uncertainty in the cluster sector; improvement here would be the highest-leverage target for future work.*

BAO-only. Running without the Planck H_0 prior gives $H_0 = 65.8^{+1.2}_{-1.1}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with the full posterior at 0.6σ. *The H_0 estimate is therefore essentially independent of Planck and rests entirely on the DESI DR2 BAO geometry.*

10 Discussion

10.1 Two faces of SSEE under Bayesian inference

The results reveal a sharp bifurcation:

- **Dark-energy sector**: $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.05σ of DESI DR2. The CPL free-fit posterior $(-0.80, -0.58)$ confirms dynamical dark energy. SSEE is predictively competitive here.

- **Background structure**: $\Omega_{\rm DE}^{\rm SSEE} = 0.840$ creates ΔBIC $= +218$ via the enlarged r_d. The tension is in Ω_m (21σ), not in H_0 ($< 1\sigma$).

10.2 Structural reinterpretation

The ΔBIC penalty assumes the Planck Ω_m prior is applicable to SSEE. It is not: in SSEE the KAL_0 amplification of $\rho_{\rm bar}$ modifies the effective clustering contribution, so $\Omega_m^{\rm eff} \neq \Omega_m^{\Lambda\rm CDM}$. Applying a ΛCDM-calibrated Ω_m prior to an SSEE universe constitutes a non-equivalent prior calibration. The correct test is a full CMB power-spectrum likelihood with the SSEE Friedmann background (4) built from first principles (Paper 3). Paper 3 must derive the SSEE-consistent background rather than adjusting Ω_m to fit: parameter adjustment within a ΛCDM framework would not constitute a genuine test of the structural hypothesis.

The Folding Symmetry (Section 2.4) provides a structural resolution: the 21σ tension in r_d is a diagnostic artefact of applying the unmodified Friedmann background to the SSEE universe. Equation (6) shows that once the acoustic-mapping operator $|w_0|$ is applied, $r_{d,\rm eff} \approx 147.5\,$Mpc recovers the Planck value within 1σ, reducing the residual tension from 21σ to sub-1σ. The 21σ figure is therefore not a falsification of the dynamical ansatz, but a quantitative signal that the standard Friedmann background requires modification in the acoustic regime.

10.3 Roadmap for Paper 3

Four ordered tests for SSEE to survive a full CMB analysis:

1. **Acoustic peak positions.** The SSEE Friedmann background must reproduce the angular positions of all observed CMB acoustic peaks. Failure here would immediately falsify the background geometry.

2. **Damping tail.** $\Omega_{m,\rm eff} = 0.160$ must be compatible with the Silk-damping scale and the third acoustic peak height. This is the most constraining test for the low-matter fraction.

3. **Joint posterior consistency.** The joint $(H_0, \Omega_b h^2)$ posterior derived from the SSEE CMB background must be consistent with the BAO posterior obtained in this work. Inconsistency would signal internal tension within the model.

4. **Transfer function derivation.** The semi-linear transfer function integrating the KAL_0 structural viscosity into the acoustic perturbation regime must be formally derived and validated against the full CMB power spectrum.

10.4 Falsifiability

SSEE makes specific, testable predictions. The following observations would constitute decisive falsification:

- **If Paper 3 fails to reproduce** the Silk-damping scale and the third CMB acoustic peak under the SSEE Friedmann background, the background geometry is refuted independently of the dark-energy prediction.
- **If Ly-α forest observations exclude** $\sum m_\nu = 0.085\,\mathrm{eV}$ at $> 3\sigma$, the neutrino bridge in the cluster-mass formula breaks and SSEE loses its only mechanism for reconciling $\Omega_{m,\mathrm{eff}}$ with cluster observations.
- **If future BAO surveys at** $z > 3$ **measure** $r_d < 160\,\mathrm{Mpc}$ directly, the Folding Symmetry hypothesis is ruled out and the 21σ background tension becomes a genuine falsification rather than a calibration diagnostic.

11 Conclusions

We have conducted a Bayesian MCMC validation of SSEE-V3.6 against DESI DR2, Planck 2018, and galaxy-cluster masses. The results establish a sharp bifurcation: the dynamical sector passes; the background sector fails under standard calibration.

1. The zero-free-parameter prediction $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.05σ of the DESI DR2 free-fit CPL posterior — a strong result given the absence of any fitted parameter.
2. Cluster masses achieve $\chi^2_r = 0.131$ with IGIMF; removing the IGIMF correction raises χ^2_r to 12.22 ($\Delta\chi^2 \approx +85$), confirming its essential role.
3. The fixed $\Omega_{m,\mathrm{eff}} = 0.160$ enlarges r_d by 19%, yielding $\Delta\mathrm{BIC} = +218$. The dynamic sector in isolation achieves $\Delta\mathrm{BIC} = -5.55$, demonstrating that the penalty is entirely attributable to background mismatch, not to the w_0–w_a prediction.
4. The Folding Symmetry hypothesis — $r_{d,\mathrm{eff}} = r_{d,\mathrm{SSEE}} \cdot |w_0| \approx 147.5\,\mathrm{Mpc}$ — would recover the Planck sound horizon within 2σ if confirmed, reinterpreting the 21σ tension as a diagnostic of non-equivalent background calibration rather than a falsification of the dynamical ansatz.
5. Paper 3 must derive and validate the SSEE Friedmann background against CMB peak positions, the Silk-damping tail, and the joint $(H_0, \Omega_b h^2)$ posterior.

Central finding: SSEE-V3.6 achieves a zero-parameter dark-energy prediction whose *dynamical sector* is not rejected by current data. The background geometry fails under standard Friedmann assumptions; whether this constitutes a physical falsification of the framework depends on the validity of applying ΛCDM-calibrated priors to an SSEE background, which is the subject of Paper 3.

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (`https://arxiv.org/abs/2503.14738`). The compressed Planck 2018 prior is from Planck Collaboration [2020]. Cosmic Chronometer $H(z)$ measurements are from Moresco et al. [2022]. Galaxy-cluster dynamical masses are from Zhang et al. [2026]. The SSEE analysis scripts are available at `https://github.com/mikealmeida1721/SSEE`.

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